

Homework 2 Solutions

Donald Timpson

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Problem 5.1

Proof. If x and y are odd integers, then $x + y$ is an even integer.

1. Let x and y be odd integers.
2. Then there exists two integers a and b such that $x = 2a + 1$ and $y = 2b + 1$. (Definition of odd)
3. Then $x + y = 2a + 2b + 2 = 2(a + b + 1)$. (Substitution and distribution)
4. Then there exists an integer c such that $x + y = 2c$, namely $c = a + b + 1$, showing that $2|(x + y)$. (Definition of divisible).
5. Since $2|(x + y)$, then $x + y$ is an even integer. (Definition of even)
6. Therefore $x + y$ is even. □

Problem 5.2

Proof. If x is an odd integer and y is an even, then $x + y$ is an odd integer.

1. Let x be an odd integer and let y be an even.
2. Then there exists two integers a and b such that $x = 2a + 1$ and $y = 2b$. (Definition of odd and even)
3. Then $x + y = (2a + 1) + 2b = 2a + 2b + 1 = 2(a + b) + 1$. (Substitution and distribution)
4. Then there exists an integer c such that $x + y = 2c + 1$, namely $c = a + b$.
5. Since $x + y = 2c + 1$, then $x + y$ is an odd integer. (Definition of odd)
6. Therefore $x + y$ is odd. □

Problem 5.3

Proof. If n is an odd integer, then $-n$ is also odd.

1. Let n be an odd integer.
2. Then there exists an integer a such that $n = 2a + 1$. (Definition of odd)
3. Then $-n = -(2a + 1) = -2a - 1 = -2a - 2 + 1 = 2(-a - 1) + 1$. (Substitution and distribution)
4. Then there exists an integer c such that $-n = 2c + 1$, namely $c = -a - 1$.
5. Since $-n = 2c + 1$, then $-n$ is an odd integer. (Definition of odd)
6. Therefore $-n$ is odd. □

Problem 5.4

Proof. If x and y are even integers, then $x \cdot y$ is even.

1. Let x and y be even integers.
2. Then there exists two integers a and b such that $x = 2a$ and $y = 2b$. (Definition of even)
3. Then $x \cdot y = (2a)(2b) = 2(2a \cdot b)$.
4. Then there exists an integer c such that $x \cdot y = 2c$, namely $c = 2a \cdot b$, showing that $2|(x \cdot y)$. (Definition of divisible)
5. Since $2|(x \cdot y)$, then $x \cdot y$ is an even integer. (Definition of even)
6. Therefore $x \cdot y$ is even. □

Problem 5.5

Proof. If x is an odd integer and y is an even, then $x \cdot y$ is an even integer.

1. Let x be an odd integer and let y be an even.
2. Then there exists two integers a and b such that $x = 2a + 1$ and $y = 2b$. (Definition of odd and even)
3. Then $x \cdot y = (2a + 1)(2b) = 4ab + 2b = 2(2ab + b)$. (Substitution and distribution)
4. Then there exists an integer c such that $x \cdot y = 2c$, namely $c = 2ab + b$, showing that $2|(x \cdot y)$. (Definition of divisible)
5. Since $2|(x \cdot y)$, then $x \cdot y$ is an even integer. (Definition of even)
6. Therefore $x \cdot y$ is even. □

Problem 5.6

Proof. If x and y are odd integers, then $x \cdot y$ is an odd integer.

1. Let x and y be odd integers.
2. Then there exists two integers a and b such that $x = 2a + 1$ and $y = 2b + 1$. (Definition of odd)
3. Then $x \cdot y = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$. (Substitution and distribution)
4. Then there exists an integer c such that $x \cdot y = 2c + 1$, namely $c = 2ab + a + b$.
5. Since $x \cdot y = 2c + 1$, then $x \cdot y$ is an odd integer. (Definition of odd)
6. Therefore $x \cdot y$ is an odd integer. □

Problem 5.7

Proof. If x is an odd integer, then x^2 is an odd integer.

1. Let x be an odd integer.
2. Then there exists an integer a such that $x = 2a + 1$. (Definition of odd)
3. Then $x^2 = (2a + 1)(2a + 1) = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1$. (Substitution and distribution)
4. Then there exists an integer c such that $x^2 = 2c + 1$, namely $c = 2a^2 + 2a$.
5. Since $x^2 = 2c + 1$, then x^2 is an odd integer. (Definition of odd)
6. Therefore x^2 is an odd integer. □

Problem 5.8

Proof. If x is an odd integer, then x^3 is an odd integer.

1. Let x be an odd integer.
2. Then there exists an integer a such that $x = 2a + 1$. (Definition of odd)
3. Then $x^3 = (2a + 1)(2a + 1)(2a + 1) = 8a^3 + 12a^2 + 6a + 1 = 2(4a^3 + 6a^2 + 3a) + 1$. (Substitution and distribution)
4. Then there exists an integer c such that $x^3 = 2c + 1$, namely $c = 4a^3 + 6a^2 + 3a$.
5. Since $x^3 = 2c + 1$, then x^3 is an odd integer. (Definition of odd)
6. Therefore x^3 is an odd integer. □

Problem 5.9

Proof. If a , b , and c are integers such that $a|b$ and $a|c$, then $a|(b + c)$.

1. Let a , b , and c be integers such that $a|b$ and $a|c$.
2. Then there exists two integers x and y such that $a \cdot x = b$ and $a \cdot y = c$. (Definition of divisible)
3. Then $b + c = a \cdot x + a \cdot y = a(x + y)$. (Substitution and distribution)
4. Then there exists an integer z such that $a \cdot z = (b + c)$, namely $z = x + y$. (Substitution)
5. Since $a \cdot z = (b + c)$, then $a|(b + c)$. (Definition of divisible)
6. Therefore $a|(b + c)$. □

Problem 5.10

Proof. If a, b , and c are integers such that $a|b$, then $a|(b \cdot c)$.

1. Let a, b , and c be integers and let $a|b$.
2. Then there exists an integer x such that $a \cdot x = b$. (Definition of divisible)
3. Then $b \cdot c = a \cdot x \cdot c = a(x \cdot c)$. (Substitution and distribution)
4. Then there exists an integer z such that $a \cdot z = (b \cdot c)$, namely $z = x \cdot c$. (Substitution)
5. Since $a \cdot z = b \cdot c$, then $a|(b \cdot c)$. (Definition of divisible)

□

Problem 5.11

Proof. If a, b, d, x , and y are integers such that $d|a$ and $d|b$, then $d|(ax + by)$.

1. Let a, b, d, x , and y be integers and let $d|a$ and $d|b$.
2. Then there exists integers j and k such that $d \cdot j = a$ and $d \cdot k = b$. (Definition of divisible)
3. Then $ax + by = (dj)x + (dk)y = d(jx + ky)$. (Substitution and distribution)
4. Then there exists an integer z such that $d \cdot z = ax + by$, namely $z = jx + ky$. (Substitution)
5. Since $a \cdot z = (ax + by)$, then $a|(ax + by)$. (Definition of divisible)

□

Problem 5.12

Proof. If a, b, c , and d are integers such that $a|b$ and $c|d$, then $(ac)|(bd)$.

1. Let a, b, c , and d be integers such that $a|b$ and $c|d$.
2. Then there exists integers x and y such that $ax = b$ and $cy = d$. (Definition of divisible)
3. Then $bd = (ax)(cy) = (ac)(xy)$. (Substitution and distribution)
4. Then there exists an integer z such that $(ac) \cdot z = bd$, namely $z = xy$. (Substitution)
5. Since $(ac) \cdot z = bd$, then $(ac)|(bd)$. (Definition of divisible)

□

Problem 5.13

Proof. The integer x is odd if and only if $x + 1$ is even.

1. ($\Rightarrow$) Let x be an odd integer.
2. Then there exists an integer a such that $x = 2a + 1$.
3. Then $x + 1 = 2a + 1 + 1 = 2a + 2 = 2(a + 1)$.
4. Then there exists an integer b such that $x + 1 = 2b$, namely $b = a + 1$.
5. Therefore $x + 1$ is even.
6. ($\Leftarrow$) Let $x + 1$ be even.
7. Then there exists an integer a such that $x + 1 = 2a$.
8. Then $x = 2a - 1 = 2a - 2 + 1 = 2(a - 1) + 1$.
9. Then there exists an integer b such that $x = 2b + 1$, namely $b = a - 1$.
10. Therefore x is odd.

□

Problem 5.18

Proof. If n is an integer, then $(n + 1)^2 - n^2$ is odd.

1. Let $n \in \mathbb{Z}$.
2. Then $(n + 1)^2 - n^2 = (n^2 + 2n + 1) - n^2 = 2n + 1$.
3. Then $\exists c \in \mathbb{Z}$ s.t. $(n + 1)^2 - n^2 = 2c + 1$, namely $c = n$.
4. Therefore $(n + 1)^2 - n^2$ is odd. (Definition of odd).

□

Problem 5.22

Proof. An integer is odd if and only if it is the sum of two consecutive integers.

1. ($\Rightarrow$) Let $n \in \mathbb{Z}$ s.t. n is odd.
2. Then $\exists x \in \mathbb{Z}, n = 2x + 1$. (Definition of odd)
3. Then $n = 2x + 1 = x + (x + 1)$, which is the sum of two consecutive integers.
4. Therefore if n is odd, then it is the sum of two consecutive integers.
5. ($\Leftarrow$) Let $x \in \mathbb{Z}$, and observe that x and $x + 1$ are consecutive integers.
6. Let $n = x + (x + 1)$, and observe that $n = x + (x + 1) = 2x + 1$ is odd. (Definition of odd)
7. Therefore the sum of two consecutive integers is odd. □

Problem 6.1

Proof. If a and b are integers with $a|b$, then $a \leq b$ is not always true.

1. Let $a = 5$, and $b = -5$.
2. There exists some integer c such that $ac = b$, namely $c = -1$, showing that $a|b$. (Definition of divisible)
3. However, $a \leq b$ is not true. □

Problem 6.3

Proof. If a , b , and c are positive integers with $a|(b \cdot c)$, then $a|b$ or $a|c$ is not always true.

1. Let $a = 6$, $b = 2$ and $c = 3$.
2. There exists some integer d such that $a \cdot d = b \cdot c$, namely $d = 1$, showing that $a|(bc)$. (Definition of divisible)
3. Observe that the only integer factors of b are 1, -1, 2, and -2, and the only integer factors of c are 1, -1, 3, and -3.
4. Because a is not a factor of either b or c , there exists no integer x such that $ax = b$ or $ax = c$.
5. Therefore both $a|b$ and $a|c$ are false. (Definition of divisible) □

Problem 6.5

Proof. If p and q are prime numbers, then $p + q$ is not always composite.

1. Let $p = 2$ and $q = 3$, then $p + q = 5$.
2. There does not exist an integer c such that $1 < c < 5$ and $c|5$.
3. Therefore $p + q$ is not a composite number. (Definition of composite) □

Problem 6.11

Proof. An integer x is positive if and only if $x + 1$ is positive is not true.

1. Let the integer $x = 0$.
2. Then $x + 1 = 1$.
3. Observe that 0 is not positive while 1 is positive. □

Problem 7.1

- a) False
- b) True
- c) False
- d) False
- e) True

Problem 7.2

Proof. $\wedge$ and $\vee$ are commutative

x	y	$x \wedge y$	$y \wedge x$	$x \vee y$	$y \vee x$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

□

Problem 7.3

Proof. 7.3

x	y	$x \wedge y$	$x \wedge \neg y$	$x \wedge y \vee x \wedge \neg y$
T	T	T	F	T
T	F	F	T	T
F	T	F	F	F
F	F	F	F	F

□

Problem 7.4

Proof. 7.4

x	y	$x \Rightarrow y$	$\neg y \Rightarrow \neg x$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

□

Problem 7.5

Proof. 7.5

x	y	$x \Leftrightarrow y$	$\neg y \Leftrightarrow \neg x$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T

□

Problem 7.6

Proof. 7.6

x	y	$x \Rightarrow y$	$y \Rightarrow x$	$x \Rightarrow y \wedge y \Rightarrow x$	$x \Leftrightarrow y$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

□

Problem 7.7

Proof. 7.7

x	y	$\neg x$	$\neg y$	$x \Rightarrow y$	$\neg x \Rightarrow \neg y$	$(x \Rightarrow y) \wedge (\neg x \Rightarrow \neg y)$	$x \Leftrightarrow y$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	F	F
F	F	T	T	T	T	T	T

□

Problem 7.11

To show that a Boolean expression is a tautology using a truth table, construct the table and demonstrate that the expression's column's values are always *true* for every possible combination.

- a) *Proof.* The expression $(x \vee y) \vee (x \vee \neg y)$ is a tautology.

x	y	$x \vee y$	$x \vee \neg y$	$(x \vee y) \vee (x \vee \neg y)$
T	T	T	T	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

□

- b) *Proof.* The expression $(x \wedge (x \Rightarrow y)) \Rightarrow y$ is a tautology.

x	y	$x \Rightarrow y$	$x \wedge (x \Rightarrow y)$	$(x \wedge (x \Rightarrow y)) \Rightarrow y$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

□

- c) *Proof.* The expression $(\neg(\neg x)) \Leftrightarrow x$ is a tautology.

x	$\neg x$	$\neg(\neg x)$	$(\neg(\neg x)) \Leftrightarrow x$
T	F	T	T
F	T	F	T

□

- d) *Proof.* The expression $x \Rightarrow x$ is a tautology.

x	$x \Rightarrow x$
T	T
F	T

□

- e) *Proof.* The expression $((x \Rightarrow y) \wedge (y \Rightarrow z)) \Rightarrow (x \Rightarrow z)$ is a tautology.

x	y	z	$((x \Rightarrow y) \wedge (y \Rightarrow z)) \Rightarrow (x \Rightarrow z)$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	T

□

- f) *Proof.* The expression $F \Rightarrow x$ is a tautology.

x	$F \Rightarrow x$
T	T
F	T

□

- g) *Proof.* The expression $(x \Rightarrow F) \Rightarrow \neg x$ is a tautology.

x	$x \Rightarrow F$	$(x \Rightarrow F) \Rightarrow \neg x$
T	F	T
F	T	T

□

- h) *Proof.* The expression $(x \Rightarrow y) \wedge (x \Rightarrow \neg y) \Rightarrow \neg x$ is a tautology.

x	y	$(x \Rightarrow y) \wedge (x \Rightarrow \neg y)$	$(x \Rightarrow y) \wedge (x \Rightarrow \neg y) \Rightarrow \neg x$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	T

□

Problem 7.13

a) *Proof.* The expression $(x \vee y) \wedge (x \vee \neg y) \wedge \neg x$ is a contradiction.

x	y	$\neg x$	$\neg y$	$x \vee y$	$x \vee \neg y$	$(x \vee y) \wedge (x \vee \neg y)$	$(x \vee y) \wedge (x \vee \neg y) \wedge \neg x$
T	T	F	F	T	T	T	F
T	F	F	T	T	T	T	F
F	T	T	F	T	F	F	F
F	F	T	T	F	T	F	F

□

b) *Proof.* The expression $x \wedge (x \Rightarrow y) \wedge \neg y$ is a contradiction.

x	y	$\neg x$	$\neg y$	$x \Rightarrow y$	$x \wedge (x \Rightarrow y)$	$x \wedge (x \Rightarrow y) \wedge \neg y$
T	T	F	F	T	T	F
T	F	F	T	F	F	F
F	T	T	F	T	F	F
F	F	T	T	T	F	F

□

c) *Proof.* The expression $(x \Rightarrow y) \wedge (\neg x \Rightarrow y) \wedge \neg y$ is a contradiction.

x	y	$\neg x$	$\neg y$	$x \Rightarrow y$	$\neg x \Rightarrow y$	$(x \Rightarrow y) \wedge (\neg x \Rightarrow y)$	$(x \Rightarrow y) \wedge (\neg x \Rightarrow y) \wedge \neg y$
T	T	F	F	T	T	T	F
T	F	F	T	F	T	F	F
F	T	T	F	T	T	T	F
F	F	T	T	T	F	F	F

□